Crypt-Arithmetic Problem

AI-4.py - C:/Users/Vineela/OneDrive/Documents/AI-4.py (3.15.0)

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```
from itertools import permutations

letters = "SENDMORY"

for digits in permutations(range(10), len(letters)):
    d = dict(zip(letters, digits))

    if d["S"] == 0 or d["M"] == 0:
        continue

    SEND = 1000*d["S"] + 100*d["E"] + 10*d["N"] + d["D"]
    MORE = 1000*d["M"] + 100*d["O"] + 10*d["R"] + d["E"]
    MONEY = (10000*d["M"] + 1000*d["O"] +
             100*d["N"] + 10*d["E"] + d["Y"])

    if SEND + MORE == MONEY:
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        break
```

IDLE Shell 3.15.0

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```
Python 3.15.0 (tags/v3.15.0:d220409, Oct  9 2026, 11:55:56) [MSC v.1951 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============== RESTART: C:/Users/Vineela/OneDrive/Documents/AI-4.py ==============
SEND = 9567
MORE = 1085
MONEY = 10652
>>>
```